3D HOSPITAL INTERIOR NAVIGATION AND VISUALIZATION

Perspective Projection + Z-Buffer hidden-surface rendering, built from scratch in Python

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1 Abstract & Objective

Objective: To develop a 3D hospital environment supporting realistic navigation and depth rendering.

Outcome: An interactive hospital model with correct perspective and hidden-surface rendering.

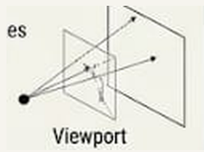
2 Technique 1: Perspective Projection

$$x_s = c_x + f \cdot X/Z$$

$$y_s = c_y - f \cdot Y/Z$$

$$f = (W/2) \,/\, tan(FOV/2)$$

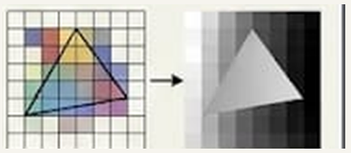
Let x_s, y_s be screen coordinates, f focal length, X, Y, Z be 3D camera-space coordinates



3 Technique 2: Z-Buffer (Depth Buffer)

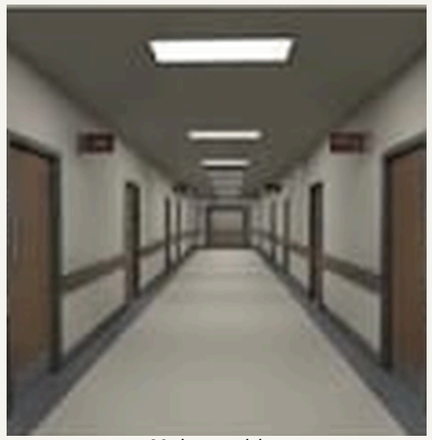
$$1/z = \lambda_0/z_0 + \lambda_1/z_1 + \lambda_2/z_2$$

Hyperbolic interpolation over a triangle





Lobby & waiting area

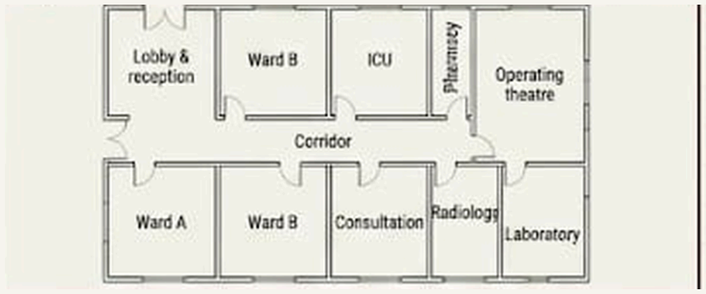


Main corridor

4 Rendering Pipeline (per frame)

Hospital model 3,966 triangles	**View transform** camera pos/yaw/pitch	**Cull + near clip** back faces, z < 0.08
Perspective projection	**Rasterise + Z-buffer test**	**Shade, fog, display**

5 Hospital Floor Plan (40 x 24m)



6 Features & Key Figures

- **10 rooms:** lobby & reception, corridor, Ward A/B, ICU, operating theatre, pharmacy, consultation, radiology, laboratory
- **~4,000 triangles**
- **Free navigation:** WASD + mouse/arrow look
- **Teaching modes:** Z-buffer ON/OFF, depth-map view, hidden-line wireframe
- **~10 fps** at 480x300 on one CPU core
- **Stack:** Python 3, NumPy, Pygame, Pillow

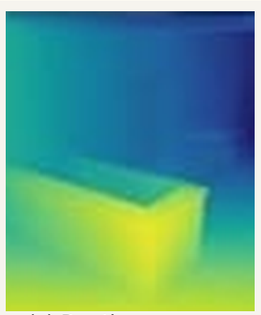
7 Results: Why the Z-Buffer Matters (same camera, same model)



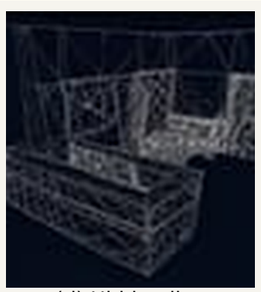
(a) Z-buffer OFF: wrong, the laboratory shows through walls



(b) Z-buffer ON: correct occlusion



(c) Depth map: near yellow, far blue



(d) Hidden-line wireframe of the triangle mesh



Ward



Operating theatre



Radiology

Conclusion & Future Work

Sums up project success in software-only real-time rendering. Future optimization likes such as spatial partitioning (BSP trees) and basic shadic shading & lighting.